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1 2 3 4 5 6 7 8 9 10
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1) PURE STATE

$$\rho = |\psi\rangle\langle\psi|$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}|2+\rangle + \frac{1}{\sqrt{2}}|2-\rangle$$

$$\text{Tr}[\rho] = 1$$

✓ 3

$$\left(\frac{1}{\sqrt{2}}|2+\rangle + \frac{1}{\sqrt{2}}|2-\rangle\right)\left(\frac{1}{\sqrt{2}}\langle 2+| + \frac{1}{\sqrt{2}}\langle 2-|\right)$$

$$\frac{1}{2} + \frac{1}{2} = 1$$

$$\frac{1}{\sqrt{2}}|\uparrow\downarrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\uparrow\rangle$$

$$\frac{1}{\sqrt{2}}|2+\rangle |2-\rangle$$

$$|\psi-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

$$|\psi+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

$$\frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle + |\uparrow\downarrow\rangle) = |\uparrow\uparrow\rangle \otimes \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$$

$\underbrace{\quad}_{|\psi\rangle} \otimes \underbrace{\quad}_{|\phi\rangle}$

$$\textcircled{5} \frac{1}{2}(|\uparrow\uparrow\rangle - |\downarrow\downarrow\rangle - |\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

$$\frac{1}{\sqrt{2}}[|\uparrow\uparrow\rangle \otimes (|\uparrow\rangle - |\downarrow\rangle)] + \frac{1}{\sqrt{2}}[|\downarrow\downarrow\rangle \otimes (|\uparrow\rangle - |\downarrow\rangle)]$$

$$= \frac{1}{2}(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle) \otimes \underbrace{(|\uparrow\rangle - |\downarrow\rangle)}_{\text{same}}$$

QUESTION 3

$$\langle \psi | \phi \rangle = 0$$

$$\frac{1}{\sqrt{2}} \langle 0 |$$

TEST 1

$$\frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle \quad \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle$$

1

$$\left(\frac{1}{\sqrt{2}} \langle 0 | - \frac{1}{\sqrt{2}} \langle 1 | \right) \left(\frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right) = \frac{1}{2} - \frac{1}{2} = 0$$

TEST 5

$$|0\rangle - |1\rangle \quad |0\rangle + |1\rangle$$

$$(\langle 0 | - \langle 1 |) (|0\rangle + |1\rangle) = 1 - 1 = 0$$

$$|1\rangle^2 + |1\rangle^2 \neq 1$$

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$$\langle a | a \rangle \langle b | b \rangle \geq |\langle a | b \rangle|^2$$

$$[A, B] = iK$$

$$\begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 0 & i \\ -i & 2 \end{pmatrix} = \begin{pmatrix} 0-2i & i+4 \\ 0 & 2i \end{pmatrix}$$

$$= \begin{pmatrix} -2i & 4+i \\ 0 & 2i \end{pmatrix} = i \begin{pmatrix} -2 & -4+i \\ 0 & 2 \end{pmatrix}$$

$$\Delta A \Delta B \geq \frac{1}{2} |\langle K \rangle|, [A, B] = iK$$

$$[A, B] = AB - BA$$

$$|\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 0 & i \\ -i & 2 \end{pmatrix} = \begin{pmatrix} -2i & 4+i \\ 0 & 2i \end{pmatrix}$$

$$BA = \begin{pmatrix} 0 & i \\ -i & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 2i & 0 \\ 4-i & -2i \end{pmatrix}$$

$$[A, B] = AB - BA = \begin{pmatrix} -2i & 4+i \\ 0 & 2i \end{pmatrix} - \begin{pmatrix} 2i & 0 \\ 4-i & -2i \end{pmatrix} = \begin{pmatrix} -4i & 4+i \\ i-4 & 4i \end{pmatrix}$$

$$\langle \psi | [A, B] | \psi \rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} -4i & 4+i \\ i-4 & 4i \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = -4i$$

$$\Delta A \Delta B \geq \frac{1}{2} |-4i| = 2$$

Suppose A and B are two observables on a quantum system and let K be the observable such that:

$$[A, B] = iK$$

Then, for any quantum state $|\psi\rangle$,

$$\Delta A \Delta B \geq \frac{1}{2} |\langle K \rangle|$$

When I measure A and B there are some states with which the uncertainty is the smallest possible and there are other states where the product of uncertainty is larger.

$$\langle \psi | K | \psi \rangle =$$

$$= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} -2 & -4+i \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} -2 \\ 0 \end{pmatrix} = -2$$

$$|\langle K \rangle| = 2$$

5) QUESTION

$$|\phi\rangle = \frac{1}{2} |0_A\rangle |0_B\rangle + \frac{\sqrt{3}}{2} |0_A\rangle |1_B\rangle$$

$$\begin{pmatrix} |0_B\rangle & |1_B\rangle \\ \langle 0_B| \\ \langle 1_B| \end{pmatrix}$$

$$\langle 0_B | |\phi\rangle | 0_B \rangle =$$

$$\langle 0_B | \left(\frac{1}{2} |0_A\rangle |0_B\rangle + \frac{\sqrt{3}}{2} |0_A\rangle |1_B\rangle \right) | 0_B \rangle$$

CHECK TRACE
OF RESULT IT
SHOULD BE 1

$$|\phi\rangle = \frac{1}{2} |0_A 0_B\rangle + \frac{\sqrt{3}}{2} |0_A 1_B\rangle$$

$$\rho = |\phi\rangle \langle \phi| = \frac{1}{4} |0_A 0_B\rangle \langle 0_A 0_B| + \frac{3}{4} |0_A 1_B\rangle \langle 0_A 1_B| + \frac{\sqrt{3}}{4} |0_A 0_B\rangle \langle 1_A 0_B| + \frac{\sqrt{3}}{4} |0_A 1_B\rangle \langle 0_A 0_B|$$

$$\text{Then } \rho_A = \text{Tr}_B[\rho] = \frac{1}{4} |0_A\rangle \langle 0_A| \underbrace{\text{Tr}(|0_B\rangle \langle 0_B|)}_{=1} + \frac{3}{4} |0_A\rangle \langle 0_A| \underbrace{\text{Tr}(|1_B\rangle \langle 1_B|)}_{=1} + \frac{\sqrt{3}}{4} |0_A\rangle \langle 0_A| \underbrace{\text{Tr}(|0_B\rangle \langle 1_B|)}_{=0} + \frac{\sqrt{3}}{4} |0_A\rangle \langle 0_A| \underbrace{\text{Tr}(|1_B\rangle \langle 0_B|)}_{=0}$$

Partial Trace

$$= \frac{1}{4} |0\rangle \langle 0| + \frac{3}{4} |0\rangle \langle 0| + 0 |0\rangle \langle 0| + 0 |0\rangle \langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

6) QUESTION

$$|\psi(\epsilon)\rangle = U(\epsilon) |\psi(0)\rangle = e^{-i/\hbar H \epsilon} |\psi(0)\rangle$$

$$2\hbar\omega, \hbar\omega$$

$$|2+\rangle, |2-\rangle$$

$$|\psi(0)\rangle = |x+\rangle = \frac{1}{\sqrt{2}} (|2+\rangle + |2-\rangle)$$

$$|\psi(\epsilon)\rangle = \frac{1}{\sqrt{2}} \left[e^{-i/\hbar 2\hbar\omega \epsilon} |2+\rangle + e^{-i/\hbar \hbar\omega \epsilon} |2-\rangle \right]$$

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$$O = 3|0x0\rangle - |1x1\rangle + (2+i)|0x1\rangle + (2-i)|1x0\rangle$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\langle O^2 \rangle = \langle \psi | O | \psi \rangle = \langle \psi | (3|0x0\rangle - |1x1\rangle + (2+i)|0x1\rangle + (2-i)|1x0\rangle) | \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \rangle$$

$$= \frac{1}{\sqrt{2}} \langle \psi | (3|0\rangle + |1\rangle - (2+i)|0\rangle + (2-i)|1\rangle)$$

$$= \frac{1}{2} (\langle 0| - \langle 1|) | (3|0\rangle + |1\rangle - (2+i)|0\rangle + (2-i)|1\rangle)$$

$$= \frac{1}{2} (3 - 1 - (2+i) - (2-i)) =$$

$$= \frac{1}{2} (2 - 2 - 1 - 2 + 1) = \boxed{-1}$$

Handwritten notes for the first calculation:

$$3|0x0\rangle \cdot \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$3 \cdot \frac{1}{\sqrt{2}} |0x0|0\rangle - 3 \cdot \frac{1}{\sqrt{2}} |0x0|1\rangle$$

Arrows point to $\langle 0|0\rangle = 1$ and $\langle 0|1\rangle = 0$.

TRICK

$$\langle O^2 \rangle = \langle O^2 \rangle - \langle O \rangle^2 \quad |\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\langle \psi | O | \psi \rangle = \dots = \boxed{-1} \Rightarrow \langle O \rangle^2 = (-1)^2 = 1$$

$$O^2 = (3|0x0\rangle \dots) (3|0\rangle \dots) = \langle \psi | O^2 | \psi \rangle = 6 \Rightarrow \langle O^2 \rangle = 6 - 1 = \boxed{5}$$

QUESTION

THE SUM MUST BE 4

$$M = \begin{pmatrix} 1 & i & 0 \\ -i & 2 & -i \\ 0 & i & 1 \end{pmatrix}$$

Eigenvalues: $\{ \dots, \dots, ? \}$

ONE ELEMENT IS 0

$$\begin{pmatrix} 1-\lambda & i & 0 \\ -i & 2-\lambda & -i \\ 0 & i & 1-\lambda \end{pmatrix}$$

$$(1-\lambda) \det \begin{pmatrix} 2-\lambda & -i \\ i & 1-\lambda \end{pmatrix} + i \det \begin{pmatrix} -i & -i \\ 0 & 1-\lambda \end{pmatrix} + 0$$

$$(2-\lambda)(1-\lambda) + i^2$$

$$2 - \lambda^2 - \lambda - 2\lambda + 1$$

$$-\lambda^2 - 3\lambda + 3$$

$$(1-\lambda)(-\lambda^2 - 3\lambda + 3)$$

